

Generation Interconnection System Impact Study Report

For Queue Project Q21-1967

Doyle Energy Facility 100 kV

May 04, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1967. The Interconnection Customer proposes a 154.0 MW battery energy storage facility interconnecting at the Doyle Energy Facility 100 kV substation in the Michigan area. The request was received on December 28, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 614 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$29,367,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: COPUS-HERNDON 100kV
Contingency: FERMI 500/230 kV Transformer
Loading: 117.6% (Rating: 213 MVA)
DFAX: 39.5%
Pre-project loading: 89.0%
2. Facility: COPUS-LEWISVILLE SWITCH 100kV
Contingency: FERMI 500/230kV Autotransformer
Loading: 114.6% (Rating: 201 MVA)
DFAX: 39.5%
Pre-project loading: 84.2%
3. Facility: COUNTY LINE 19-MIDWA M.P 100kV
Contingency: FERMI 500 kV/230 kV Transformer Bank
Loading: 114.3% (Rating: 407 MVA)
DFAX: 60.5%
Pre-project loading: 91.4%
4. Facility: COUNTY LINE 19-S WOOROW 100kV
Contingency: FERMI 500-230 kV Autotransformer
Loading: 108.1% (Rating: 498 MVA)
DFAX: 60.5%
Pre-project loading: 89.4%
5. Facility: BONITA SPRINGS-MIDWAY .P SUBSTATION 100kV L
Contingency: FERMI SUBSTATION 500/230 kV Autotransformer
Loading: 105.1% (Rating: 425 MVA)
DFAX: 60.5%
Pre-project loading: 83.2%
6. Facility: LANDSMAN CRK-SAMARIA 100kV Line 2
Contingency: SAMARIA-LANDSMAN CREEK 100kV #1
Loading: 101.9% (Rating: 191 MVA)

DFAX: 14.0%

Pre-project loading: 90.7%

7. Facility: MONROE MI-MONROE 100kV

Contingency: FERMI SUBSTATION 500/230kV TX

Loading: 101.9% (Rating: 211 MVA)

DFAX: 12.5%

Pre-project loading: 92.7%

8. Facility: DUARTE-ST IGNACE 100kV Circuit 1

Contingency: FERMI 500-230 kV Transformer

Loading: 100.6% (Rating: 184 MVA)

DFAX: 16.2%

Pre-project loading: 87.1%

9. Facility: SAMARIA-ST IGNACE 100kV

Contingency: FERMI 500-230 kV TX

Loading: 100.4% (Rating: 191 MVA)

DFAX: 16.2%

Pre-project loading: 87.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
COPUS-HERNDON 100kV	Rebuild line with higher-capacity conductor	\$5,752,000	100% (\$5,752,000)
COPUS-LEWISVILLE SWITCH 100kV	Rebuild line with higher-capacity conductor	\$11,344,000	100% (\$11,344,000)
COUNTY LINE 19-MIDWAY M.P 100kV Line	Rebuild line with higher-capacity conductor	\$2,795,000	100% (\$2,795,000)
COUNTY LINE 19-S WOOROW 100kV	Reconductor line	\$1,645,000	100% (\$1,645,000)
BONITA SPRINGS SUBSTATION-MIDWAY M.P SUBSTATION 100kV	Reconductor line	\$934,000	100% (\$934,000)
LANDSMAN CRK-SAMARA SUBSTATION 100kV	Reconductor line	\$1,072,000	100% (\$1,072,000)
MONROE MI-MONROE 100kV	Reconductor line	\$1,143,000	100% (\$1,143,000)
ST IGNACE-DUARTE 100kV Circuit 1	Reconductor line	\$1,773,000	100% (\$1,773,000)
SAMARIA-ST IGNACE 100kV	Reconductor line	\$2,909,000	100% (\$2,909,000)
	Total allocated to project		\$29,367,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.